

Skander Moalla

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EDUCATION

EPFL

Lausanne, Switzerland

PhD in Computer and Communication Sciences Sep 2022 – Present (graduation expected in Fall 2026)

Deep reinforcement learning (RL) and LLM post-training. Advised by Prof. Caglar Gulcehre (CLAIRE).

University of Oxford

Oxford, UK

MSc in Advanced Computer Science

Oct 2021 – Sep 2022

MSc thesis in multi-agent reinforcement learning with Prof. Shimon Whiteson's group (WhiRL).

École Polytechnique

Paris, France

BSc in Mathematics and Computer Science (Double Major)

Sep 2017 – Aug 2020

Graduated Summa Cum Laude | CGPA: 4.09/4.0 | Awarded an excellence scholarship.

PUBLICATIONS

Simon Matrenok*, **Skander Moalla***, and Caglar Gulcehre. Quantile Reward Policy Optimization: Alignment with Pointwise Regression and Exact Partition Functions. *NeurIPS*, 2025.

Project Apertus, [...], **Skander Moalla***, [...], Antoine Bosselut, Martin Jaggi, and Imanol Schlag. Apertus: Democratizing open and compliant llms for global language environments. *arXiv*, 2025.

Skander Moalla, Andrea Miele, Razvan Pascanu, and Caglar Gulcehre. No Representation, No Trust: Connecting Representation, Collapse, and Trust Issues in PPO. *NeurIPS*, 2024.

Xiuying Wei, **Skander Moalla**, Razvan Pascanu, and Caglar Gulcehre. Building on Efficient Foundations: Effectively Training LLMs with Structured Feedforward Layers. *NeurIPS*, 2024.

Skander Moalla*, Manuel Madeira*, Lorenzo Riccio*, and Joonhyung Lee*. [Re] Reproducibility Study of Behavior Transformers. *ReScience C*, 9(2), 2023. **Outstanding Paper Awards Honorable Mentions.**

Benjamin Ellis, Jonathan Cook, **Skander Moalla**, Mikayel Samvelyan, Mingfei Sun, Anuj Mahajan, Jakob Foerster, and Shimon Whiteson. SMACv2: An Improved Benchmark for Cooperative Multi-Agent Reinforcement Learning. *NeurIPS*, 2023.

EXPERIENCE

Google DeepMind

Paris, France

PhD Student Researcher

Sep 2025 – Dec 2025

- Hosted by Alexandre Ramé in the Gemma post-training team (Olivier Bachem, Lucas Dixon).
- Worked on leveraging diversity to improve test-time scaling (pass@k) and sampling in reinforcement learning.
- Contributed to the Gemma open-source JAX codebase (Gemma 3 KV cache sharing and sampler).

Quincus

Singapore (Remote)

Applied Research Intern

Mar 2021 – Aug 2021

- Deployed a solution leveraging deep reinforcement learning (RL) to optimize middle-mile logistics.
- Built simulators to model multi-mile parcel shipping and trained deep RL prototypes (PPO, DQN).
- Stack: Python (OpenAI Gym & Baselines, LightGBM, pytest), Docker, NSCC High-Performance Computing.

Amazon

Software Development Intern

Luxembourg

Sep 2020 – Feb 2021

- Completed full-stack and DevOps sprint tasks within a “two-pizza team” owning multiple services end-to-end.
- Designed and built a prototype to detect similar and duplicate business metrics supporting Amazon’s transportation network worldwide to ease metric discovery and reduce computation costs leveraging deep learning and NLP techniques with AWS SageMaker.
- Stack: AWS, Python, Java (Dagger, AutoValue, JUnit, Mockito), TypeScript (ImmutableJS, Mocha, Chai).

EPFL – Laboratory for Topology and Neuroscience & Blue Brain Project

Geneva, Switzerland

Research Intern

Jan 2020 – Apr 2020

- BSc thesis supervised by Prof. Kathryn Hess Bellwald. (Link to thesis).
- Built a framework for neuron simulation and leveraged tools from information theory and algebraic topology (simplicial homology) to formulate new conclusions about the dynamics of neurons forming directed cliques.
- Stack: Python 3 (DIT, JupyterLab, NumPy, SciPy, Brian2), C++.

Ooredoo

Tunis, Tunisia

Full-Stack Developer Intern

Jul 2018 – Aug 2018

- Implemented a PoC for a custom notification system in a hybrid mobile app featuring profiling functionalities and allowing Ooredoo to adjust the activity of their notification server. Stack: Ionic 3, Node.js, socket.io.

TEACHING

Université Dauphine Tunis (Université Paris Dauphine-PSL)

Tunis, Tunisia

Visiting Lecturer – Reinforcement Learning

Jan 2026

- Delivered an intensive 24-hour course for the Master’s in Big Data & Artificial Intelligence (BDIA).
- Covered reinforcement learning foundations and modern LLM post-training methods such as RLHF.
- Integrated theoretical lectures with hands-on labs and a final project implementing a deep Actor-Critic.

EPFL

Lausanne, Switzerland

LauzHack Deep Learning Bootcamp Instructor (Deep Reinforcement Learning)

Jul 2024

- Delivered a 2-hour introductory lecture on deep reinforcement learning (Bellman equations, DQN, PPO, etc.).

EPFL

Lausanne, Switzerland

Teaching Assistant & Supervisor

Sep 2022 – Present

- Supervised 9 MSc projects and theses.
- Probability and Statistics (MATH-232 Spring 2023), Machine Learning (CS-433 Fall 2023 – exam preparation), Artificial Neural Networks / Reinforcement Learning (CS-456 Spring 2024 – project preparation).

ATSM – Tunisian Association of Mathematics

Tunis, Tunisia

Olympiad Coach

Jul 2018 – Present

- Designed material for Olympiad problem-solving and trained national mathematics team members for the Mediterranean Mathematical Olympiad (MYMC 2018) and the International Mathematical Olympiad (IMO 2020).

SOFTWARE

Core stack and responsibilities.....

Co-lead the Swiss AI post-training efforts

July 2024 – Present

- Developed infrastructure and offline post-training code to train the Swiss open models Apertus 8B and 70B on the CSCS Alps cluster with 1000+ GH200 nodes.

Reference for infrastructure & compute platforms in the lab

Sep 2023 – Present

- Support the deployment of container-based research experiments on multiple clusters at the scale of hundreds of nodes, including Slurm HPC clusters and Run:AI (Kubernetes-based) clusters. (Documentation)
- Stack: Docker, Kubernetes, Run:AI, Slurm, NVIDIA Pyxis, NVIDIA Enroot, Apptainer/Singularity.

Open-source contributions and selected projects.....

Open-source contributions

- @PyTorch/RL, @DLR-RM/stable-baselines3, @Farama-Foundation/Gymnasium, @HuggingFace/TRL.

Python Machine Learning Research Project Template

Sep 2022 – Present

- Maintain a template for reproducible Python machine learning research projects deployable on various platforms including Slurm (with containers through NVIDIA Pyxis/Enroot or Apptainer) and Kubernetes clusters.

Class projects.....

SAM - Study Abroad Matching

Sep 2019 – Dec 2019

- Designed a web app around an algorithm to match students applying to study-abroad programs and universities based on a human-in-the-loop solution to the Gale-Shapley college admissions problem.
- Stack: Python, Django, React, Google Cloud Platform (GCP), Google App Engine, Google Cloud SQL.

Computer Facial Discrimination - Machine Learning Project

May 2019 – Jun 2019

- Studied the effect of different pre-trained feature extractors (Autoencoders, CNN classifiers) and clustering methods on the discrimination of human faces.
- Built a customizable autoencoder model to demonstrate that training data (as an analogy for background and culture) can be the root of multiple kinds of discrimination.
- Stack: Python, TensorFlow, Keras, scikit-learn, Google Colab, Kaggle.

TOP5 Basketball Team Manager - C++ Project

Nov 2018 – Jan 2019

- Managed a team of 9 students / Led the Game Engine team / Maintained the Git repository.
- Designed and implemented the game engine and its interaction with the remaining game entities.
- Stack: C++, Qt.

Meet Halfway - Web Project

May 2018 – Jun 2018

- Built a user-friendly website that allows a group of friends to find the best destination to travel to and meet for holidays according to their respective locations and cost of travel.
- Stack: HTML, CSS, JS.

EDUCATION - EXCHANGE PROGRAMS

University of Toronto

Toronto, Canada

Exchange – Fall 2019 | Software Engineering, Databases, Compilers & Interpreters

Sep 2019 – Dec 2019

Stanford University

Stanford, USA

Exchange – Summer 2019 | GPA: 4.15/4.0 | AI, Entrepreneurship & Innovation

Jun 2019 – Aug 2019

LANGUAGES

English: Bilingual French: Native Arabic: Native German: Elementary